PAUL-CHRISTIAN BÜRKNER

GENERAL INFORMATION

Date of Birth	16 June 1991
Place of Birth	Marburg, Germany
Work Address	Vogelpothsweg 87, 44227 Dortmund, Germany
Email	paul.buerkner@gmail.com
Website	https://paulbuerkner.com

KEY SCIENTIFIC METRICS

Publications	148 (peer-reviewed only)
Funding	2,053,000 € (third-party only)
Citations	39,999 (source: GoogleScholar)
h-index	54 (source: GoogleScholar)

WORK EXPERIENCE

since 2023 — **Full Professor of Computational Statistics**

TU Dortmund University

Full Professor of Computational Statistics, Department of Statistics, TU Dortmund University, Germany.

2020-2023 — **Independent Junior Research Group Leader**

University of Stuttgart

Independent Junior Research Group Leader for Bayesian Statistics at the Cluster of Excellence SimTech, University of Stuttgart, Germany.

2019-2020 — **Postdoctoral Researcher**

Aalto University

Postdoctoral researcher at the chair of Computational Probabilistic Modeling (Prof. Dr. Vehtari), Aalto University, Department of Computer Science, Finland.

2014-2019 — **Research Associate**

University of Münster

Research associate at the chair of Statistics and Methods (Prof. Dr. Holling), University of Münster, Department of Psychology, Germany.

HIGHER EDUCATION

2014-2017 — **University of Münster**

PhD in Psychology

Grade: Summa Cum Laude · Institute of Psychology
Title: *Optimal Design and Bayesian Data Analysis*.
Received multiple awards (see the Awards section).

2014-2017 — **University of Hagen**

Master of Mathematics

Grade: 1.3 · Institute of Mathematics
Thesis: *On the Statistics of Curie–Weiss–Distributed Random Variables*.

2013-2014 — **University of Münster**

Master of Psychology

Grade: 1.1 · Institute of Psychology
Thesis: *Adaptive Designs for Logistic Models with False Answers*.

2011-2014 — **University of Hagen**

Bachelor of Mathematics

Grade: 1.7 · Institute of Mathematics
Thesis: *A Hull Operator for Complex Matroids*.

2010-2013 — **University of Münster**

Bachelor of Psychology

Grade: 1.2 · Institute of Psychology
Thesis: *Testing for Publication Bias in Diagnostic Meta-Analysis: A Simulation Study*.

RESEARCH FUNDING

GACR & DFG — Modrak M. & Bürkner P. C. (2026). grammo: A grammar of probabilistic models. *Funder: Czech Research Foundation (GACR) and German Research Foundation (DFG)*. Total: 556,000 €. Own share: **338,000 €**.

MERCUR — Bürkner P. C., Hoffmann D., & Kobeleva X. (2026). Deep Bayes for Biomedicine. *Funder: MERCUR Foundation*. Total: 300,000 €. Own share: **100,000 €**.

DFG — Fried et al. (2024). Spatio-temporal Statistics for the Transition of Energy and Transport (Collaborative Research Center). *Funder: German Research Foundation (DFG)*. Total: 12,540,000 €. Own share: **312,000 €**.

DFG — Bürkner P. C. & Radev S. T. (2023). BayesFlow: Simulation Intelligence with Deep Learning. *Funder: German Research Foundation (DFG)*. **353,000 €**.

DFG — Bürkner P. C. (2022). Intuitive Joint Priors for Bayesian Multilevel Models. *Funder: German Research Foundation (DFG)*. **238,000 €**.

DFG — Bürkner P. C. (2022). Bayesian Distributional Latent Variable Models. *Funder: German Research Foundation (DFG)*. **238,000 €**.

DFG — Bürkner P. C. & Bulling A. (2022). Amortized Bayesian Inference for Multilevel Models. *Funder: German Research Foundation (DFG)*. **232,000 €**.

EXC SimTech — Guthke A. & Bürkner P. C. (2022). Data-Integrated Training of Surrogate Models for Uncertainty Quantification and Diagnostics of Complex Biological Systems. *Funder: Cluster of Excellence SimTech*. **285,000 €**.

Cyber Valley — Bürkner P. C. (2021). Meta-Uncertainty in Bayesian Model Comparison. *Funder: Cyber Valley Research Fund*. **242,000 €**.

EXC SimTech — Bürkner P. C. & Sedlmair M. (2021). Machine Learning for Bayesian Model Building. *Funder: Cluster of Excellence SimTech*. **285,000 €**.

EXC SimTech — Bulling A. & Bürkner P. C. (2021). Bayesian Intent Prediction for Human-Machine Collaboration. *Funder: Cluster of Excellence SimTech*. **175,000 €**.

ELLIS — Bulling A. Bürkner P. C., Kuchenbecker J. K., Pradel M., Schulte im Walde S., Staab S., Steinwart I., & Vu T. (2021). Stuttgart ELLIS Unit. *Funders: ELLIS Society and University of Stuttgart*.

SELECTED AWARDS

GCPR — 2023 · **Best paper honorable mention award** at the German Conference of Pattern Recognition (GCPR).

SIPS — 2020 · **Mission award** of the Society for Improving Psychological Science (SIPS) for brms.

SIPS — 2020 · **Commendation award** of the Society for Improving Psychological Science (SIPS) for brms.

University of Münster — 2018 · **Award for the best dissertation** 2017-2018 in Psychology at the University of Münster.

German Society for Psychology — 2017 · Gustav A. Lienert **Award for the best methodological dissertation** in Psychology awarded by the German Society for Psychology (DGPs).

University of Münster — 2017 · **Award for the best lecture** at the Institute of Psychology in Münster.

German National Acad. Foundation — 2014 · **Scholarship** of the German National Academic Foundation (Studienstiftung des deutschen Volkes).

OPEN-SOURCE SOFTWARE

brms — Lead author · An R package for Bayesian regression models using Stan. Received multiple awards (see the Awards section).

posterior — Lead author · An R package for working with posterior distributions.

thurstonianIRT — Lead author · An R Package for fitting Thurstonian IRT models.

BayesFlow	Author · A Python library for simulation-based Bayesian inference.
posteriordb	Author · A Posterior Database for Bayesian Inference.
elicito	Author · A Python package for expert prior elicitation.
loo	Author · An R package for approximate leave-one-out cross-validation.
ggsimplex	Author · An R package for simplex visualizations with ggplot2.
bayesim	Author · An R package for simulations with Bayesian models.
bayehear	Author · An R package for metrics to evaluate Bayesian models.
bayesfam	Author · An R package for custom brms families.
bayesian	Author · An R package to interface brms and tidymodels.
rstan	Contributor · An R Interface to Stan.
bayesplot	Contributor · An R package for visualizing Bayesian models.
projpred	Contributor · An R package for projection predictive variable selection.
emmeans	Contributor · An R package for estimating marginal means.

SELECTED PROFESSORSHIP CALLS

Full Professorship	2022 · Call for the Full Professorship (W3) in Computational Statistics, Department of Statistics, TU Dortmund University, Germany. **Accepted**.
Full Professorship	2022 · Call for the Full Professorship (W3) in Data Analytics and Computational Statistics, Department of Computer Science, University of Konstanz, Germany.

SELECTED SCIENTIFIC INVOLVEMENT

Scholar	since 2026 · Scholar of the ELLIS Society in the Program Robust Machine Learning (https://ellis.eu/research/programs/robust-machine-learning).
Area Chair	since 2026 · Area chair of NeurIPS and ICML.
Principle Investigator	since 2026 · Principle investigator of the Ruhr Innovation Lab (excellence university association). Involved in preparing the on-site visit.
Associate Member	since 2026 · Associate member of the Horst Görtz Institute for IT Security (https://hgi.rub.de/).
Associate Member	since 2026 · Associate member of the ELLIS Unit NRW.
Editor	since 2025 · Editor of the Journal of Robustness Reports.
Member	since 2024 · Member of the comittee for the re-accreditation of Data Science studies at TU Dortmund University.
Founding Member	2024 · Founding member of the Computational Clinical Psychology and Psychotherapy Network (https://ccpp.network/) funded by the DFG.
Chairman	2024 · Chairman of the appointment committee for the associate professorship in Causality at TU Dortmund University.
Reviewer	since 2024 · Reviewer in appointment procedures for professorships.
Organizer	2022 · Organizer of the 1st International SimTech Summer School, University of Stuttgart. Co-Organizers: Benjamin Unger and Kristyna Pluhackova.
Reviewer	since 2022 · Reviewer for major funding agencies. *Selection*: German Research Foundation (DFG), European Research Council (ERC), Dutch Research Council (NWO), Humboldt Foundation.
Member	since 2021 · Member of the ELLIS Society (https://ellis.eu/).
Member	2021 – 2024 · Member of Cyber-Valley (https://cyber-valley.de/en).
Faculty Member	2021 – 2023 · Faculty Member of the International Max Planck Research School for Intelligent Systems (IMPRS-IS; https://imprs.is.mpg.de/).
Founding Member	2021 – 2023 · Founding member of the Stuttgart ELLIS Unit (https://ellis.eu/units/stuttgart).

Member	since 2018 · Member of the Stan Development Team (https://mc-stan.org/).
Consultant	since 2018 · Academic consultant in industry. Selection: Bayer (2018), Novartis (since 2021), Axem (since 2022).
Editor	2018 – 2020 · Associate editor of Biostatistics.
Reviewer	since 2014 · Reviewer for international journals and conferences. *Selection*: AISTATS, Bayesian Analysis, Behavior Research Methods, Biometrical Journal, ICML, ICLR, Journal of Machine Learning Research, Journal of Probability and Statistics, Journal of Statistical Software, Journal of the American Statistical Association, Journal of the Royal Statistical Society, Nature, Nature Ecology & Evolution, Nature Human Behaviour, NeurIPS, Philosophical Transactions, Psychological Methods, Psychometrika, Psychonomic Bulletin and Review, Statistics in Medicine.

SELECTED TALKS

Multilevel Conference	2026 · Utrecht · Keynote Title: *Amortized Bayesian Inference for Multilevel Models*.
QUAMU Conference	2026 · Bielefeld · Invited Talk Title: *Bayes in the Age of Machine Learning*.
ELLIS Unconference	2025 · Copenhagen · Invited Talk Title: *Robust Amortized Bayesian Inference with Self-Consistency Losses*.
Bayescomp	2025 · Singapore · Contributed Talk Title: *Robust Amortized Bayesian Inference with Self-Consistency Losses*.
One World ABI Seminar	2025 · online · Invited Talk Title: *Amortized Mixture and Multilevel Models*.
Cyber Valley	2025 · Stuttgart · Invited Talk Title: *Meta-Uncertainty in Bayesian Model Comparison*.
University of Bielefeld	2025 · online · Invited Talk Title: *The future of Bayes and brms with application in biology*.
Stan Conference	2024 · Oxford · Contributed Talk Title: *Generative Bayesian Modeling with Implicit Priors*.
PHYSTAT-SBI Workshop	2024 · Munich · Invited Talk Title: *A Statistical Perspective on Simulation-Based Inference*.
Bayes on the Beach Conference	2024 · Gold Coast · Keynote Title: *Does Bayes have to be slow? A glimpse into amortized Bayesian inference*.
Oxford University	2023 · Oxford · Keynote Title: *Probabilistic Modeling for Ecology*.
Princeton University	2023 · online · Invited Talk Title: *An Introduction to Bayesian Statistics*.
DagStat Conference	2022 · Hamburg · Contributed Talk Title: *The sparse polynomial chaos expansion: a fully Bayesian approach with joint priors on the coefficients and global selection of terms*.
Psychoco Conference	2021 · online · Keynote Title: *Bayesian Item Response Models*.
Oslo UseR Group	2021 · online · Invited Talk Title: *Bayesian multilevel modeling with brms*.
Oxford University	2020 · online · Invited Talk Title: *Bayesian regression modeling*.
Turku University	2020 · online · Invited Talk Title: *Bayesian multilevel modeling with brms*.
TU Dortmund University	2020 · online · Invited Talk Title: *Bayesian multilevel modeling with brms*.
Bayer	2020 · online · Invited Talk Title: *Bayesian multilevel modeling with brms*.

Stat. Methods for Linguistics	2019 · Potsdam · Keynote Title: *A Principled Bayesian Workflow for Data Analysis*.
University of Duisburg-Essen	2019 · Essen · Invited Talk Title: *A Principled Bayesian Workflow for Data Analysis*.
DGPs Conference	2019 · Kiel · Contributed Talk Title: *Improving Convergence Diagnostics for MCMC Sampling Algorithms*.
Stan Conference	2019 · Cambridge · Contributed Talk Title: *Leave-Future-Out Cross-Validation for Bayesian Time-Series Models*.
Multilevel Conference	2019 · Utrecht · Keynote Title: *Bayesian Multilevel Modeling with brms and Stan*.
DagStat	2019 · Munich · Contributed Talk Title: *Leave-Future-Out Cross-Validation for Bayesian Time-Series Models*.
Stan Conference	2018 · Helsinki · Contributed Talk Title: *Custom Response Distributions in brms*.
EAM Conference	2018 · Jena · Contributed Talk Title: *Handling Ordinal Predictors in Regression Models via Monotonic Effects*.
Bayes@Lund	2018 · Lund · Keynote Title: *Why Not to be Afraid of Priors*.
DGPs Conference	2017 · Tübingen · Keynote Title: *Optimal Design and Bayesian Data Analysis*.
eRum Conference	2016 · Poznan · Contributed Talk Title: *brms: An R Package for Bayesian Multilevel Models using Stan*.
Int. Workshop on Simulation	2015 · Vienna · Contributed Talk Title: *Adaptive Designs for Logistic Models with False Answers*.
DGPs Conference	2015 · Jena · Contributed Talk Title: *Optimal Design of Non-Parametric Two-Sample Tests*.

SELECTED WORKSHOPS

Multilevel Conference	2026 · Utrecht · 1 day Title: *Bayesian modeling with brms*.
Oxford University	2023 · Department of Biology · 1 day Title: *Bayesian modeling for biologists using brms*.
University of Tübingen	2023 · Center of Methods · 2 days Title: *Bayesian modeling with the brms package*.
TU Dortmund University	2022 · Department of Statistics · 2 days Title: *Bayesian Statistics*.
University of Salzburg	2022 · Department of Psychology · 2 days Title: *Introduction to Bayesian Data Analysis*.
Oxford University	2021 · Department of Zoology · 4 days Title: *Bayesian Regression Modelling for Biologists*.
Research Cluster SMiP	2020 · Mannheim · 2 days Title: *Introduction to Stan: A Probabilistic Programming Language for Bayesian Inference*.
University of Aarhus	2020 · Department of Economics and Business Economics · 1 day Title: *Bayesian Model and Variable Selection*.
MPI for Human Development	2019 · Göttingen · 1 day Title: *Bayesian Multilevel Modeling*.
MPI for Emp. Aesthetics	2019 · Frankfurt · 2 days Title: *Bayesian Multilevel Modeling*.
Multilevel Conference	2019 · Utrecht · 1 day Title: *Introduction to Bayesian Data Analysis*.

DagStat Conference
2019 · Munich · 1 day
Title: *Bayesian Data Analysis using Stan*.

University of Lausanne
2018 · Department of Psychology · 2 days
Title: *Introduction to Meta-Analysis*.

University of Magdeburg
2018 · Department of Psychology · 4 days
Title: *Introducing Basic and Advanced Bayesian Modelling*.

University of Aarhus
2018 · 4 days
Title: *Advanced Bayesian Statistical Modeling*.

ETH Zurich
2018 · 1 day
Title: *Classical and Bayesian Multi-Level Models in R*.

University of Hamburg
2017 · Department of Psychology · 2 days
Title: *Fitting Multi-Level Models in R*.

DPPD Conference
2017 · Munich · 1 day
Title: *Bayesian Multi-Level Models in R with brms*.

University of Bern
2017 · Department of Psychology · 3 days
Title: *Bayesian Multi-Level Models in R with brms*.

University of Münster
2017 · Department of Psychology · 3 days
Title: *Introduction to Bayesian Inference*.

University Paris Decardes
2017 · 1 day
Title: *Introduction to Meta-Analysis*.

SELECTED TEACHING ACTIVITIES

Data Science and Statistics
2025–2026 · TU Dortmund University · 2 times
Lecture: *Simulation-Based Inference*.

Data Science and Statistics
2024–2026 · TU Dortmund University · 2 times
Lecture: *Statistical Learning for Big Data*.

Data Science and Statistics
2024–2026 · TU Dortmund University · 3 times
Lecture: *Case Studies I and II*.

Data Science and Statistics
2024-2026 · TU Dortmund University · 3 times
Seminar: *Model Comparison*.

Data Science and Statistics
2023–2026 · TU Dortmund University · 3 times
Lecture: *Applied Bayesian Data Analysis*.

Data Science and Statistics
2023-2026 · TU Dortmund University · 4 times
Seminar: *Multilevel Models*.

Data Science and Statistics
2023 · TU Dortmund University · 1 times
Lecture: *Computational Statistics*.

Simulation Science
2022 · University of Stuttgart · 1 times
Lecture: *Bayesian Statistics and Probabilisitic Machine Learning*.

Simulation Science
2021 · University of Stuttgart · 1 times
Lecture: *ML Sessions: Bayesian Statistics*.

Simulation Science
2021-2022 · University of Stuttgart · 2 times
Seminar: *Advanced Topics in Simulation Science*.

Psychology
2018 · University of Münster · 2 times
Seminar: *Structural Equation Modeling and Bayesian Statistics*.
Average Evaluation: 10.9 points (15 point *abitur* scale).

Psychology
2014-2019 · University of Münster · 5 times
Lecture: *Descriptive Statistics and Probability Theory*.
Average Evaluation: 12.6 points (15 point *abitur* scale).
Award for the best lecture in the winter semester 2016/2017.

Psychology
2015-2018 · University of Münster · 4 times
Lecture: *Inferential Statistics*.
Average Evaluation: 12.1 points (15 point *abitur* scale).

CURRENT PHD STUDENTS

TU Dortmund University
since 2026 · Yiming Zang · Statistics
Topic: *Deep Bayes for Biomedicine*.

TU Dortmund University
since 2026 · Hans Olischläger · Statistics
Topic: *Improving Neural Architectures for Amortized Bayesian Inference*.

TU Dortmund University
since 2024 · Svenja Jedhoff · Statistics
Topic: *Real-time Spatio-Temporal Data Analysis for Monitoring Logistics Networks*.
Co-Advisor: Anne Meyer

TU Dortmund University
since 2024 · Aayush Mishra · Statistics
Topic: *Robust and Efficient Learning in Amortized Bayesian Inference*.

TU Dortmund University
since 2024 · Lars Kühmichel · Statistics
Topic: *BayesFlow: Simulation Intelligence with Deep Learning*.
Co-Advisor: Stefan Radev

GRADUATED PHD STUDENTS

TU Dortmund University
2022 – 2026 · Florence Bockting · Statistics
Topic: *Simulation-based expert prior elicitation: Method and software development*.

TU Dortmund University
2022 – 2026 · Luna Fazio · Statistics
Topic: *Extending the modeling toolkit of psychological sciences with Bayesian distributional latent variable models*.

University of Tübingen
2022 – 2026 · Soham Mukherjee · Statistics
Topic: *On Latent Variable Estimation using Derivative Gaussian Processes*.
Co-Advisor: Manfred Claassen

TU Dortmund University
2021 – 2025 · Javier Aguilar · Statistics
Topic: *Intuitive Joint Priors for High-Dimensional Bayesian Linear Models*.

University of Stuttgart
2022 – 2025 · Philipp Reiser · Computer Science
Topic: *Bayesian Surrogate Modeling for Uncertainty-Aware Inference and Data Integration*. Co-Advisor: Anneli Guthke

University of Stuttgart
2021 – 2025 · Marvin Schmitt · Computer Science
Topic: *Towards Trustworthy Amortized Bayesian Inference with Deep Learning*.

University of Stuttgart
2021 – 2025 · Maximilian Scholz · Computer Science
Topic: *Simulation-Assisted Workflows for Bayesian Model Building*.

Aalto University
2019 – 2023 · Alejandro Catalania · Computer Science
Topic: *Robust Bayesian Inference: variable and structure selection and variational inference*. Primary Advisor: Aki Vehtari

University of Münster
2018 – 2021 · Niklas Schulte · Psychology
Topic: *Statistical Properties of Forced-Choice Questionnaires in Applicant Personality Measurements*. Primary Advisor: Heinz Holling

CURRENT POSTDOCTORAL RESEARCHERS

TU Dortmund University
since 2024 · Šimon Kucharský
Topic: *Applications of Amortized Bayesian Inference*.

TU Dortmund University
since 2023 · Daniel Habermann
Topic: *Amortized Bayesian Inference for Multilevel Models*.

FORMER POSTDOCTORAL RESEARCHERS

University of Stuttgart
2022 – 2024 · Lei Shi
Topic: *Bayesian Intent Prediction for Human-Machine Collaboration*.
Co-Advisor: Prof. Andreas Bulling

University of Heidelberg
2021 – 2023 · Stefan Radev
Topic: *Amortized Bayesian Inference*.
Became an assistant professor at Rensselaer Polytechnic Institute, Troy, USA.

ALL PUBLICATIONS

In Review

180) Aguilar J. E. & **Bürkner P. C.** (in review). Dependency-Aware Shrinkage Priors for High Dimensional Regression. *ArXiv preprint*.

179) Aguilar J. E., Kohns D., Vehtari A., & **Bürkner P. C.** (in review). R2 priors for Grouped Variance Decomposition in High-dimensional Regression. *ArXiv preprint*.

178) Bignardi G., Kievit R. A., & **Bürkner P. C.** (in review). A general method for estimating reliability using Bayesian Measurement Uncertainty. *PsyArXiv preprint*.

177) Bockting F. & **Bürkner P. C.** (in review). elicito: A Python Package for Expert Prior Elicitation. *ArXiv preprint*.

176) Fazio, L., & **Bürkner, P. C.** (in review). Latent Variable Models for Distributional Features. *ArXiv preprint*.

175) Fazio L., Scholz M., & **Bürkner P. C.** (in review). Primed Priors for Simulation-Based Validation of Bayesian Models. *ArXiv preprint*.

174) Habermann, D., Bulling, A., Radev, S. T., & **Bürkner, P. C.** (in review). Amortized Bayesian Inference on Multilevel Models of Arbitrary Structure. *ArXiv preprint*.

173) Kucharský Š., Mishra A., Habermann D., Radev S. T., & **Bürkner P. C.** (in review). Improving the Accuracy of Amortized Model Comparison with Self-Consistency. *ArXiv preprint*.

172) Kühmichel, L., Radev, S. T., Koppolu, B. P., Davoudi, M., Huang, J. M., & **Bürkner, P. C.** (in review). High-Dimensional Simulation-Based Inference in Latent Spaces. *ArXiv preprint*.

171) Kühmichel, L., Huang, J. M., Pratz, V., Arruda, J., Olischläger, H., Habermann, D., Kucharský, Š., Elsemüller, L., Mishra, A., Bracher, N., Jedhoff, S., Schmitt, M., **Bürkner, P. C.**, & Radev, S. T. (in review). BayesFlow 2.0: Multi-Backend Amortized Bayesian Inference in Python. *ArXiv preprint*.

170) Mishra, A., Kucharský, Š., & **Bürkner, P. C.** (in review). Unsupervised Continual Learning for Amortized Bayesian Inference. *ArXiv preprint*.

169) Modrák M., Stroppel S., & **Bürkner P. C.** (in review). Simulation-based validation of Bayes factor computation. *ArXiv preprint*.

168) Mukherjee S., Aguilar J. E., Zago M., Claassen M., & **Bürkner P. C.** (in review). Latent variable estimation with composite Hilbert space Gaussian processes. *ArXiv preprint*.

167) Nalborczyk, L. & **Bürkner, P. C.** (in review). Precise temporal localisation of M/EEG effects with Bayesian generalised additive multilevel models. *biorXiv preprint*.

166) Olischläger, H., Jedhoff, S., Kucharský, Š., Mishra, A., Radev, S. T., & **Bürkner, P. C.** (in review). Mitigating Representation Gaps in Amortized Bayesian Inference with Auxiliary Supervision. *ArXiv preprint*.

165) Riha, A. E., Jedhoff, S., **Bürkner, P. C.**, & Vehtari, A. (in review). Approximating Bayesian Leave-One-Group-out Cross-Validation. *ArXiv preprint*.

164) Riha, A. E., Lindgren, L., Kohns, D., **Bürkner, P. C.**, & Vehtari, A. (in review). To select or not to select: predictively consistent priors instead of model selection. *arXiv preprint*.

163) Schulze, L., Bohländer, J., **Bürkner, P. C.**, Röpke, S., Renneberg, B., & Zetsche, U. (in review). Negligible associations between cognitive control of negative information and daily emotion regulation in borderline personality disorder and major depression. *PsyArXiv preprint*.

162) Stevenson, N., **Bürkner, P. C.**, Heathcote, A., & Radev, S. T. (in review). Semi-Amortized Bayesian Inference with Reusable Local Posteriors. *ArXiv preprint*.

161) Walter, V., Herold, J., Feuchter, S. A., Thomae, S., Venohr, M., Vogelsberg, A., Kilic, M., Berner, F., Nanz, L., Leiter-Stoeppke, U., Sinnberg, T., Schuerch, C. M., Ulmer, A., **Bürkner, P. C.**, & Flatz, L. (in review). High immune receptor clonality in melanoma-draining lymph nodes associates with immune dysfunction and poor survival. *bioRxiv preprint*.

160) Zago, M., Mukherjee, S., Schleicher, J. T., **Bürkner, P. C.**, Tabatabai, G., & Claassen, M. (in review). G-LATO: Inference of Spatial Latent Ordering via Deep Gaussian Processes. *bioRxiv preprint*.

Accepted

159) Jedhoff S., Kutabi H., Meyer A. & **Bürkner P. C.** (accepted). Efficient Uncertainty Propagation in Bayesian Two-Step Procedures. *Statistics and Computing*.

2026

158) Bracher N., Kühmichel L., Ivanova D. R., Intes X., **Bürkner P. C.**, & Radev S. T. (2026). JADAI: Jointly Amortizing Adaptive Design and Bayesian Inference. *Proceedings of the International Conference on Machine Learning*.

157) **Bürkner, P. C.**, Gabry, J., Kay, M., & Vehtari, A. (in review). posterior: Tools for Working with Posterior Distributions in R. *Journal of Open Source Software*. doi:10.21105/joss.10526

156) **Bürkner P. C.**, Schmitt M., & Radev S. T. (2026). Simulations in Statistical Workflows. *Philosophical Transactions A*. doi:10.1098/rsta.2024.0616

155) Gelman, A., Vehtari, A., McElreath, R., Simpson, D., Margossian, C. C., Yao, Y., Kennedy, L., Gabry, J., **Bürkner, P. C.**, Modrák, M., & Leos Barajas, V. (2026). Bayesian Workflow. *Chapman & Hall/CRC*.

154) Hu N., **Bürkner P. C.**, & Arvaniti A. (2026). A tutorial on Bayesian multivariate linear mixed-effect regression using brms. *Journal of Speech, Language, and Hearing Research*. doi:10.1044/2026_JSLHR-25-00880

153) Jedhoff S., Semenova E., Raulo A., Meyer A., & **Bürkner P. C.** (2026). From Mice to Trains: Amortized Bayesian Inference on Graph Data. *Transactions in Machine Learning Research*.

152) Kallioinen, N., Paananen, T., **Bürkner, P. C.**, & Vehtari, A. (2026). priorsense: Efficient prior and likelihood sensitivity checks for Bayesian models in R. *Journal of Open Source Software*. doi:10.21105/joss.11036

151) Kołczyńska M., & **Bürkner P. C.** (2026). Does political trust strengthen democracy? *International Political Science Review*. doi:10.1177/01925121251405191

150) Kucharský Š. & **Bürkner P. C.** (2026). Amortized Bayesian Mixture Models. *Statistics and Computing*. doi:10.1007/s11222-026-10911-y

149) Li C., Vehtari A., **Bürkner P. C.**, Radev S. T., Acerbi L., & Schmitt M. (2026). Amortized Bayesian Workflow. *Transactions in Machine Learning Research*.

148) Mishra A., Habermann D., Schmitt M., Radev S. T., & **Bürkner P. C.** (2026). Robust Amortized Bayesian Inference with Self-Consistency Losses on Unlabeled Data. *International Conference on Learning Representations*.

147) Mukherjee S., Claassen M., & **Bürkner P. C.** (2026). Hilbert space methods for approximating multi-output latent variable Gaussian processes. *Statistics and Computing*. doi:10.1007/s11222-026-10869-x

146) Revathe, T., Weidling, M. T., Utami-Atmoko, S. S., Setia, T. M., Razik, I., van Schaik, C. P., Whiten, A., **Bürkner, P. C.**, & Schuppli, C. (2026). How eight years of social and individual learning interact to shape ecological competence in orangutans. *Science*. doi:10.1126/science.adz6415

145) Reiser P., **Bürkner P. C.**, & Guthke A. (2026). Bayesian Surrogate Training on Multiple Data Sources: A Hybrid Modeling Strategy. *Statistics and Computing*. doi:10.1007/s11222-026-10906-9

144) Säilynoja T., Schmitt M., **Bürkner P. C.**, & Vehtari A. (2026). Posterior SBC: Simulation-Based Calibration Checking Conditional on Data. *Statistics and Computing*. doi:10.1007/s11222-026-10825-9

143) Scheurer S., Reiser P., Brünnette T., Nowak W., Guthke A., & **Bürkner P. C.** (2026). Uncertainty-Aware Surrogate-based Amortized Bayesian Inference for Computationally Expensive Models. *Transactions in Machine Learning Research*.

142) Zetsche, U., Bohländer, J., **Bürkner P. C.**, Röpke, S., Renneberg, B., & Schulze, L. (2026). Beyond feeling down: Expectations about and memories of daily affective experiences in major depression and borderline personality disorder. *Clinical Psychological Science*. doi:10.1177/21677026261426975

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